Fresh and Storage Onions

Onions are an excellent complement to your vegetable offering. We sow a lot of them at the beginning of the season, mainly to make bunches of fresh onions. These onions do not last as long as storage onions, but you will get a better price. In addition, there is a great demand for them and consumers love them. If you wish to sell during the cold season, don't hesitate to reserve a portion of your production plan for storage onions.

Family

- Amaryllidaceae

Cultivars - Fresh

- Eclipse
- Purplette
- Sierra Blanca
- Red long of Tropea
- Ailsa Craig
- Bianca Di Maggio

Cultivars - Storage

- Redwing
- Talon
- Rossa Di Milano
- Red Carpet
- Cortland

Crop Planning

Fresh Onions

Fresh onions are one of the most profitable crops to grow. Always popular at the market, the trick is to extend the harvest window the most you can.

Any planting program needs to factor in the fact that onions are photoperiodic plants. This means that their stages of growth evolve with day length.

To factor this in, we plan two generations transplants of fresh onions per season. In the spring, we transplant these generations two weeks apart. This allows us to spread out our harvest later in the season and have a constant supply of fresh onions.

Another trick to increase this constancy in our supply of fresh onions is to include cultivars with different days to maturity in each generation. This allows for a more evenly spread out harvest.

Storage Onions

Plan for one transplant of storage onions. Transplant as soon as the soil is ready in the spring. This timing is important as onions need the short days of spring to complete their vegetative growth before starting their bulb development in the long days of summer.

Starting seeds

- +/- 45 days before the transplantation date, sow 3-4 seeds per cell in 128 plug flats. To speed up this process, the vacuum seeder can be used.
- Place trays on heating mats and maintain soil temperature at 80 °F (27 °C) until emergence. At this step, the potting-mix must be kept moist at all times.
- After emergence, the ambient temperature in the nursery should be kept around 75 °F (24 °C) during the day and 65 °F (18 °C) at night. Onions can also be kept at much cooler temperatures, but will grow more slowly.
- At week 4, trim about 1 in (2.5 cm) of the head of the plant. We use an electric pruning shear.

- One week before transplant, trim about 1 in (2.5 cm) of the head of the plant and harden them off by exposing them to outside temperatures and reducing the water intake.
- Plants are ready to transplant when the seedlings are about 7–8 weeks old and +/- 6 in (15 cm).

Intensive Spacing

3 rows (10 in apart), spaced every 8 in (20 cm) on the row.

Fertilization

Onions will do great in beds that were previously amended with compost and should be fertilized with lots of nitrogen to start with. The amount of growth and development prior to bulbing will determine the bulb size. We use a mix of alfalfa meal (2-0-2) and pelleted chicken manure (5-3-2).

Irrigation

Irrigation Type

- Drip tapes (4 per bed) or sprinkler, although overhead irrigation can increase foliar diseases.

Irrigation Management

- Root depth: Shallow (6-24 in or 15-60 cm)
- General: Steady supply of moisture; if too dry, onions develop a strong unpleasant flavor. Avoid water on leaves to minimize downy mildew.
- Storage onions : stop irrigating the onions when the first leaves start to fall, indicating the start of maturation.

Implantation

Equipment

- Fertilization and amendments
- Power harrow
- 30 in (75 cm) bed prep rake
- Plug popper
- Dibbler (or marking tubes for the bed prep rake)
- Irrigation equipment
- Insect netting and flexible hoops (optional)

Itinerary

- To facilitate the transplantation, we prepare each bed just before the implantation (since the ground is quite soft after having been harrowed).
- Remove the tarp and if there is a lot of large debris, use the rake quickly to clean each bed.
- Uniformly apply fertilization and compost on each bed and pass with the power harrow at a depth of 2 in (5 cm).
- Mark the spacing by running a dibbler over each bed.
- Bring the well-irrigated cell trays close to the bed and use the plug popper to remove the seedlings from their cells. Since this crop is not very sensitive to transplanting, rely on speed rather than perfection in the planting process. Just make sure the top of the plug is leveled with the soil.
- 1 person will drop the seedling while 2 others transplant the plants as fast as possible. 3 people/100 ft (30 m) bed is a good ratio for efficient workflow.
- Install irrigation and water until the soil is deeply moistened.

- If you have a history of damages caused by onion flies, install hoops every 5 ft (1.5 m) and cover the crop with an anti-insect net. Place sandbags on every hoop to stabilize the net.

Weeding

- Unlike other crops grown with close spacings, onions do not create a good canopy and can't outcompete weeds very well. For this reason, plan to transplant them in clean beds (free of weeds on the previous crop). Never plant onions in beds where weeds went to seeds in the previous crop.

- A good practice is to plan for a stale seedbed before the onion transplant.

- 10 days after transplanting, use a bio-disc to weed the onions. Repeat 1 to 3 more times until onions get so big that using the bio-disc will bruise or break them.

- When the onions start to reach the 4 leaves stage, weeds can be killed by using the torch of the flame weeder. Keeping the beds clean in these first few stages of the crop growth is essential for the optimal development of the onion bulb.

- If transplanting into dirty fields, doing so in landscape fabric will be the better option.

- In all scenarios, hand weeding might be necessary in the later stages of the crop, especially if weeds have a chance to set seeds before the onions are harvested.

Plant Protection

- Fresh onions are susceptible to botrytis, downy mildew and other wilts that can quickly spread to the whole crop. To prevent this, weekly scooting is essential. If some disease is identified, weekly sprays of copper and sulphur in alternance might be necessary to mitigate the spreading. Systematically inoculating plants with *Bacillus subtilis* is an alternative strategy.

- Onion’s main pests are thrips and onion flies, which can be both controlled by covering the crop with insect netting.

Harvest

Fresh Onions

- Fresh onions are harvested with their tops on and bunched in the fields in a kneeled position.
- Fresh onions are picked when their size is between 1-2 in (2.5–5 cm). Bottoms should be white (or red), without any dead leaves—make sure to strip off any leaves that are crushed, or have signs of rip damage.
- Make bunches of +/- 7-8 onions and tie them with a rubber band. Cut the tip of the leaves with a knife to allow the bunches to be stacked in the storage bins.
- 1 person should be able to harvest at least 60 bunches/hour

Storage Onions

- Storage onions are harvested when their tops start to die off and fold onto themselves.
- Choose a sunny day to harvest; the onions need to be dry at harvest time in order to cure well afterwards.
- Place the onions delicately in the harvest bins to avoid any bruises on the bulbs.

Conditioning

Fresh Onions

- Upon arriving from the fields, the bins are transferred from the cart and placed to the left of the sinks.
- A couple of bunches are placed side by side with the roots facing the operator. The operator sprays the soil off with the pressure gun. Pile the clean bunches to the right and repeat.
- Once the clean pile of onions has reached a certain height, place the onions in a storage bin and identify the bin.
- 1 person should be able to package at least 100 bunches per hour.

Storage Onions

- Cure storage onions in the nursery for 2-3 weeks. Place all the onions on the tables in a single layer. Place fans below the tables to increase air circulation.
- Note that a room with temperature, humidity and air circulation control can also be used for this purpose.
- The curing process is done when the collars are completely dry. At that stage, all stems can be cut off.
- Place the bulbs in either mesh bags or cardboard boxes.
- Make sure to discard all onions that are not completely dried or that are damaged.
- Avoid removing any peel at that stage. Onions can be stored longer with all their peels on.

Storing

Fresh Onions

- Fresh onions are stored in a cold room kept at 35 °F (2 °C)

Storage Onions

- Storage onions are stored in a cold room kept at 43-46 °F (6-8 °C)